

KHAIRY ISLAM SHAHEEN

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AI/ML and embedded systems engineer with hands-on experience in PyTorch edge inference, HuggingFace NLP pipelines, computer vision on constrained hardware, and gaze-controlled HCI research. Built a real-time drowsiness detector (DenseNet121 + CUDA), a 3-subsystem Arduino robotics platform, and a zero-trust LLM-powered enterprise chatbot with Gemini 1.5 Flash.

EDUCATION

Egypt-Japan University of Science and Technology (E-JUST)

Oct 2023 -- Feb 2028

B.Sc. Computer Science & Engineering | cGPA: 3.2

Coursework: Data Structures (C), Advanced Programming (OOP), Computer Networks, Signals & Systems, Probability & Statistics, Electronics Engineering

EXPERIENCE

Shaheen for Import & Export

Aug 2022 -- Present

IT & Digital Operations Lead

- Built a zero-trust WhatsApp AI chatbot (FastAPI + Gemini 1.5 Flash) with environment-level access control, hardcoded keyword fence interceptor neutralizing prompt injection before LLM exposure, and stateful per-user conversation sessions
- Developed Arabic text normalization middleware (normalize_arabic() with Unicode variant mapping) enabling automated bilingual filesystem operations across 1,343 enterprise files
- Implemented pure-Python ridge regression forecasting for a Django ERP (no numpy/sklearn) -- Gaussian elimination, feature engineering, explainable ML with feature importance

Digital Egypt Pioneers Initiative (DEPI Round 4)

Jun 2026 -- Present

Cybersecurity Intern

- Simulating secure network architectures, routing protocols (OSPF, EIGRP), and defense topologies in Cisco Packet Tracer

PC Building Store

Jun 2019 -- Aug 2022

Technical Assistant

- Assembled 40+ custom PCs; diagnosed hardware issues

PROJECTS

MRL Eye State Classification

PyTorch, DenseNet121, CUDA 12.6, OpenCV

- Fine-tuned DenseNet121 for binary classification on MRL Eye Dataset -- 15 epochs, RTX 3050
- Built real-time webcam inference pipeline: Haar Cascade, temporal smoothing, async inference, alerts
- Developed 21 scripts: augmentation, retrain iterations, quantization, benchmarking

ErgoX-UI

SwiftUI, ARKit TrueDepth, iPhone 16 Pro

- Designed gaze-controlled mobile UI based on CHI'25 research
- 9 SwiftUI views with adjustable dwell-time (500-1000ms), calibration, music player
- Built 53KB HTML/JS prototype for cross-platform demo

Arduino Robotics Competition

Arduino C++, Sensors

- Maze Solver: Ultrasonic wall-following (NewPing + HC-SR04), left-hand rule traversal
- Light Seeker: LDR differential-drive with analog threshold control
- Arena Robot: Bluetooth-controlled (RemoteXY) with DHT11 + MQ gas sensing

ESP32-CAM AI Smart Stick

Embedded C, Computer Vision

- Navigation aid for visually impaired, edge-device model optimization

Customer Ordering System

FastAPI, HuggingFace API, Docker

- HuggingFace distilbert for ticket priority detection; PII redaction middleware

NMEA Serial Radar Simulator

Python, NMEA 0183

- NMEA GGA + HDT sentences, configurable COM port, threaded control

TECHNICAL SKILLS

AI/ML	PyTorch (DenseNet121, CUDA 12.6), HuggingFace, Google Gemini API, OpenCV, ridge regression
Embedded	ESP32-CAM, Arduino (RemoteXY, NewPing, DHT11, MQ, LDR, HC-SR04), NMEA 0183
Languages	Python, C++, C, Swift, Arduino C++, JavaScript, SQL
Frameworks	FastAPI, Django 6 + DRF, SwiftUI + ARKit, React 19
Tools	Docker, Git, CUDA Toolkit, Cloudflare, Figma, LaTeX